

```
import pandas as pd
df=pd.read_csv("Salary_Data.csv")
```

```
display(df)
```

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0
5	2.9	56642.0
6	3.0	60150.0
7	3.2	54445.0
8	3.2	64445.0
9	3.7	57189.0
10	3.9	63218.0
11	4.0	55794.0
12	4.0	56957.0
13	4.1	57081.0
14	4.5	61111.0
15	4.9	67938.0
16	5.1	66029.0
17	5.3	83088.0
18	5.9	81363.0
19	6.0	93940.0
20	6.8	91738.0
21	7.1	98273.0
22	7.9	101302.0
23	8.2	113812.0
24	8.7	109431.0
25	9.0	105582.0
26	9.5	116969.0
27	9.6	112635.0
28	10.3	122391.0
29	10.5	121872.0

```
print(df.shape)
print(df.head())
print(df.info())
print(df.isnull().sum())
print(df.describe())
```

```
(30, 2)
```

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0

```

<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  -
0   YearsExperience  30 non-null    float64
1   Salary          30 non-null    float64
dtypes: float64(2)
memory usage: 612.0 bytes
None
YearsExperience    0
Salary            0
dtype: int64

```

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```

x=df[['YearsExperience']]
y=df[['Salary']]
print(x)
print(y)

```

	YearsExperience
0	1.1
1	1.3
2	1.5
3	2.0
4	2.2
5	2.9
6	3.0
7	3.2
8	3.2
9	3.7
10	3.9
11	4.0
12	4.0
13	4.1
14	4.5
15	4.9
16	5.1
17	5.3
18	5.9
19	6.0
20	6.8

21	7.1
22	7.9
23	8.2
24	8.7
25	9.0
26	9.5
27	9.6
28	10.3
29	10.5

	Salary
0	39343.0
1	46205.0
2	37731.0
3	43525.0
4	39891.0
5	56642.0
6	60150.0
7	54445.0
8	64445.0
9	57189.0
10	63218.0
11	55794.0
12	56957.0
13	57081.0
14	61111.0
15	67938.0
16	66029.0
17	83088.0
18	81363.0
19	93940.0
20	91738.0
21	98273.0
22	101302.0
23	113812.0
24	109431.0
25	105582.0
26	116969.0
27	112635.0
28	122391.0
29	121872.0

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=12)
```

```
from sklearn.linear_model import LinearRegression
slr=LinearRegression()
slr.fit(x_train,y_train)# model training
slr.predict(x_test) #testing
```

```
array([[ 55675.5913984 ],
       [100372.57413579],
       [ 81352.58148158],
       [ 68038.58662364],
       [ 37606.59837691],
       [ 92764.5770741 ]])

from sklearn.metrics import r2_score
y_pred=slr.predict(x_test)
r2=r2_score(y_test,y_pred)
print(r2)

0.9429269163491424
```